

Solving $Ax = 0$: Pivot Variables, Special Solutions

Lesson 6 said what the do-nothing trades are. This lesson finds every one of them, for any menu of funds: simplify A to U and then to R , and read the rank, the pivot and free variables, and every solution off R .

PART 1 Index

Note 1 of the pair: $Ax = 0$ and the null space. (Note 2 is Lesson 8, $Ax = b$.)

1. Conditions and important concepts

13 short definitions: R , pivot, rank, pivot and free variables, special solutions, N , dimension.

2. Procedure for $Ax = 0$

The 8 steps, in order.

3. 2×2 example

Nifty and its twin ETF: one free variable.

4. 3×3 example

Nifty, bonds, balanced: one free variable. And midcap: none.

5. Bigger example: two free variables

Four funds, three rules: two special solutions, the matrix N , every solution. With a try-it box.

METHODS IN THIS LESSON

Row operations keep $N(A)$

Reduced row echelon form R

Rank

Pivot and free variables

$n - \text{rank}$

Special solutions

Nullspace matrix N

Echelon form U

PART 2 Prerequisites

Click any link to review it, then come back.

- The nullspace (Lesson 6): trades with $Ax = 0$, which change nothing.
- Elimination, 3×3 (Lesson 2) and Gauss–Jordan (Lesson 3).

THE FUNDS USED IN THE EXAMPLES

Fund	What ₹1 lakh does to (budget, return, exposure)
Nifty	(1, 12, 1)
Nifty ETF (a twin of Nifty)	(1, 12, 1)
Bonds	(1, 7, 0)
Balanced (half Nifty, half bonds)	(1, 9.5, 0.5)
Midcap	(1, 15, 1.2)

Each fund is a column of A ; each rule is a row. A trade x lists ₹ lakh into each fund (positive = buy, negative = sell).

1. Conditions and important concepts

1. What is $Ax = 0$?

$Ax = 0$ means:

Find all trades x that produce ZERO change in every rule.

Budget unchanged, return unchanged, exposure unchanged. The set of all such x is called the **null space**, $N(A)$.

2. What is R ?

R = **Reduced Row Echelon Form** (RREF): the matrix we get after completely row-reducing A . We get there in two stages:

$$A \xrightarrow{\text{clear below the pivots}} U \xrightarrow{\text{pivots to 1, clear above}} R$$

METHOD · REDUCED ROW ECHELON FORM R

Eliminate below each pivot ($A \rightarrow U$), then divide each pivot row by its pivot and clear the entries above each pivot ($U \rightarrow R$). Every pivot becomes a 1, and it is the only non-zero entry in its column.

RULE · ROW OPERATIONS KEEP THE NULL SPACE

Row operations only rewrite the rules ("return minus $12 \times$ budget"). A trade that gives zero for every old rule gives zero for every new one, and the steps can be undone. So $Ax = 0$ and $Rx = 0$ have **exactly the same solutions**. That's why we can solve with R .

IMPORTANT

R does **not** mean rank. R = Reduced Row Echelon Form. Rank = number of pivots.

3. What is a pivot?

A **pivot** is a leading 1 in R (the first non-zero entry of a row). Example: Nifty, bonds, balanced (worked out in the 3×3 example):

$$R = \begin{bmatrix} \boxed{1} & 0 & 0.5 \\ 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{Pivot 1} \rightarrow \text{column 1 (Nifty)} \\ \text{Pivot 2} \rightarrow \text{column 2 (bonds)} \end{array}$$

DEFINITION · RANK = NUMBER OF PIVOTS

Here there are 2 pivots, so $\text{rank} = 2$. In money terms: only 2 of these funds bring something new.

4. What is a pivot variable?

The variable belonging to a pivot column. Above: x_1 (Nifty) and x_2 (bonds) are pivot variables.

5. What is a free variable?

A variable whose column does **not** contain a pivot. Above: x_3 (balanced) is a free variable.

RULE · PIVOT VARIABLES AND FREE VARIABLES

Pivot column $\rightarrow$ pivot variable (decided by its row of R). Column without a pivot $\rightarrow$ free variable (you choose it).

6. What does a free variable mean?

A free variable is a variable whose value **we can choose**. The rows of R above say $x_1 + 0.5x_3 = 0$ and $x_2 + 0.5x_3 = 0$, so once we choose x_3 , the pivot variables are decided:

Choose x_3 (balanced)	Then $x_1 = -0.5x_3$ (Nifty)	And $x_2 = -0.5x_3$ (bonds)	Trade
0	0	0	$(0, 0, 0)$: do nothing
1	-0.5	-0.5	$(-0.5, -0.5, 1)$
5	-2.5	-2.5	$(-2.5, -2.5, 5)$
-3	1.5	1.5	$(1.5, 1.5, -3)$

All are possible choices. In money terms: buy some balanced, sell half that amount in each of Nifty and bonds. Nothing changes.

7. Number of free variables

RULE · FREE VARIABLES = $N - \text{RANK}$

With n variables (funds): number of free variables = $n - \text{rank}$. Above: $3 - 2 = 1$.

8. Special solutions

METHOD · SPECIAL SOLUTIONS

Give **one** free variable the value 1 and all other free variables the value 0, then solve for the pivot variables. Each free variable gives **one** special solution.

$$\boxed{\text{Number of special solutions} = \text{number of free variables}}$$

Above: $x_3 = 1$ gives $s_1 = (-0.5, -0.5, 1)$.

9. When do special solutions exist?

If the number of free variables is **more than 0**: non-zero special solutions exist (Nifty, bonds, balanced: 1 free variable).

If the number of free variables is **0**: there are no non-zero special solutions, and the only solution is

$$x = 0$$

Example: Nifty, bonds, midcap reduces to $R = I$ (3 pivots, 0 free variables), so the only do-nothing trade is doing nothing (worked in the 3×3 example).

10. Nullspace matrix N

METHOD · NULLSPACE MATRIX N

Put all special solutions side by side as **columns**: if they are $s_1, s_2, \dots, s_k$, then $N = \begin{bmatrix} s_1 & s_2 & \cdots & s_k \end{bmatrix}$.

Above there is one special solution, so N has one column: $N = \begin{bmatrix} -0.5 \\ -0.5 \\ 1 \end{bmatrix}$.

11. Important null space property

Every column of N is a null space solution. Therefore:

$$AN = 0$$

Check above, row by row of $A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix}$: budget $-0.5 - 0.5 + 1 = 0 \checkmark$; return $-6 - 3.5 + 9.5 = 0 \checkmark$; exposure $-0.5 + 0 + 0.5 = 0 \checkmark$.

12. All null space solutions

If there are two special solutions s_1 and s_2 , every null space solution is

$$x = c s_1 + d s_2$$

where c and d can be **any** numbers. (With one special solution: $x = c s_1$.)

13. Null space dimension

$$\text{Dimension of null space} = \text{number of free variables} = n - \text{rank}$$

Dimension 1: a line of trades. Dimension 2: a plane. Dimension 0: just the point $x = 0$.

2. Procedure for $Ax = 0$

Step	What to do
1	Row reduce $A \rightarrow R$ (first $A \rightarrow U$, then $U \rightarrow R$).
2	Find the pivot columns.
3	Identify the free variables (columns without a pivot).
4	For each free variable: make it = 1 and all other free variables = 0.
5	Solve for the pivot variables using the rows of R .
6	Each result is a special solution.
7	Put the special solutions side by side: $N = [s_1 \ s_2 \ \cdots]$.
8	All null space solutions are combinations of those special solutions.

3. 2×2 example

Nifty and its twin ETF, with the budget and return rules. We want $Ax = 0$:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \begin{array}{l} \leftarrow \text{budget} \\ \leftarrow \text{return} \end{array}$$

Step 1, row reduce. Row 2 $- 12 \times$ row 1: $12 - 12(1) = 0$, $12 - 12(1) = 0$. The pivot is already 1, nothing above it:

$$R = \begin{bmatrix} \boxed{1} & 1 \\ 0 & 0 \end{bmatrix}$$

Steps 2–3. Pivot: column 1. Free variable: x_2 (ETF).

Equation (row 1 of R):

$$x_1 + x_2 = 0$$

Steps 4–5. Set the free variable $x_2 = 1$. Then $x_1 + 1 = 0$, so $x_1 = -1$.

Step 6, special solution:

$$s_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \text{sell ₹1 lakh Nifty, buy ₹1 lakh ETF}$$

Steps 7–8. Nullspace matrix and every solution:

$$N = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad x = c \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Check $AN = 0$: budget $1(-1) + 1(1) = 0 \checkmark$; return $12(-1) + 12(1) = 0 \checkmark$.

4. 3×3 example

Nifty, bonds, balanced, with all three rules:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{array}$$

Step 1a, $A \rightarrow U$ (clear below the pivots):

- Row 2 $- 12 \times$ row 1: $12 - 12 = 0$, $7 - 12 = -5$, $9.5 - 12 = -2.5 \rightarrow (0, -5, -2.5)$.
- Row 3 $- 1 \times$ row 1: $1 - 1 = 0$, $0 - 1 = -1$, $0.5 - 1 = -0.5 \rightarrow (0, -1, -0.5)$.
- Row 3 $- 0.2 \times$ row 2 (multiplier $(-1)/(-5) = 0.2$): 0 , $-1 + 1 = 0$, $-0.5 + 0.5 = 0 \rightarrow (0, 0, 0)$.

METHOD · ECHELON FORM U (THE STAIRCASE)

After clearing below every pivot, each row starts further right than the row above, and all-zero rows sit at the bottom.

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 1b, $U \rightarrow R$ (pivots to 1, clear above):

- Row 2 $\div (-5)$: 0 , 1 , $-2.5/(-5) = 0.5 \rightarrow (0, 1, 0.5)$.
- Row 1 $- 1 \times$ row 2: $1 - 0 = 1$, $1 - 1 = 0$, $1 - 0.5 = 0.5 \rightarrow (1, 0, 0.5)$.

$$R = \begin{bmatrix} \boxed{1} & 0 & 0.5 \\ 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 \end{bmatrix}$$

Steps 2–3. Pivot columns: 1 and 2. Free column: 3. So x_1 = pivot (Nifty), x_2 = pivot (bonds), x_3 = free (balanced). Rank = 2; free variables = $3 - 2 = 1$.

Equations (rows of R):

$$x_1 + 0.5x_3 = 0$$

$$x_2 + 0.5x_3 = 0$$

Steps 4–5. Set $x_3 = 1$. Then $x_1 = -0.5$ and $x_2 = -0.5$.

Step 6, special solution:

$$s_1 = \begin{bmatrix} -0.5 \\ -0.5 \\ 1 \end{bmatrix} \quad \text{buy ₹1 lakh balanced, sell ₹0.5 lakh each of Nifty and bonds}$$

Steps 7–8. $N = s_1$ and every solution is $x = c s_1$ (Lesson 6's $(1, 1, -2)$ is $c = -2$).

Check $AN = 0$: budget $-0.5 - 0.5 + 1 = 0$ ✓; return $12(-0.5) + 7(-0.5) + 9.5(1) = -6 - 3.5 + 9.5 = 0$ ✓; exposure $-0.5 + 0 + 0.5 = 0$ ✓.

Same size, no free variable: Nifty, bonds, midcap

- $A \rightarrow U$: row 2 $- 12 \times$ row 1 $\rightarrow (0, -5, 3)$; row 3 $-$ row 1 $\rightarrow (0, -1, 0.2)$; row 3 $- 0.2 \times$ row 2 $\rightarrow (0, 0, 0.2 - 0.6) = (0, 0, -0.4)$.
- $U \rightarrow R$: row 3 $\div (-0.4) \rightarrow (0, 0, 1)$; row 2 $- 3 \times$ row 3 $\rightarrow (0, -5, 0)$, then $\div (-5) \rightarrow (0, 1, 0)$; row 1 $-$ row 2 $-$ row 3 $\rightarrow (1, 0, 0)$.

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad 3 \text{ pivots, 0 free variables} \Rightarrow \boxed{x = 0 \text{ only}}$$

Midcap brings something new, so no fund can be rebuilt from the others.

5. Bigger example: two free variables

Four funds (Nifty, ETF, bonds, balanced), three rules:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix}$$

Step 1a, $A \rightarrow U$:

- Row 2 $- 12 \times$ row 1: $0, 0, 7 - 12 = -5, 9.5 - 12 = -2.5 \rightarrow (0, 0, -5, -2.5)$.
- Row 3 $- 1 \times$ row 1: $0, 0, 0 - 1 = -1, 0.5 - 1 = -0.5 \rightarrow (0, 0, -1, -0.5)$.
- Column 2 has nothing left to use as a pivot (the ETF column was cleared along with Nifty), so skip it.
- Row 3 $- 0.2 \times$ row 2: $0, 0, -1 + 1 = 0, -0.5 + 0.5 = 0 \rightarrow (0, 0, 0, 0)$.

Step 1b, $U \rightarrow R$:

- Row 2 $\div (-5)$: $(0, 0, 1, 0.5)$.
- Row 1 $- 1 \times$ row 2: $1, 1, 1 - 1 = 0, 1 - 0.5 = 0.5 \rightarrow (1, 1, 0, 0.5)$.

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 \\ 0 & 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Steps 2-3. Pivot variables: x_1 (Nifty), x_3 (bonds). Free variables: x_2 (ETF), x_4 (balanced). Rank = 2; free variables = $4 - 2 = 2$.

Equations:

$$x_1 + x_2 + 0.5x_4 = 0$$

$$x_3 + 0.5x_4 = 0$$

Special solution 1

$x_2 = 1, x_4 = 0$. Then $x_1 + 1 + 0 = 0 \Rightarrow x_1 = -1$, and $x_3 + 0 = 0 \Rightarrow x_3 = 0$.

$$s_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

swap ₹1 lakh Nifty for the ETF

Special solution 2

$x_2 = 0, x_4 = 1$. Then $x_1 + 0 + 0.5 = 0 \Rightarrow x_1 = -0.5$, and $x_3 + 0.5 = 0 \Rightarrow x_3 = -0.5$.

$$s_2 = \begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix}$$

build ₹1 lakh balanced from Nifty and bonds

Nullspace matrix

$$N = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \\ 0 & -0.5 \\ 0 & 1 \end{bmatrix} \begin{matrix} \leftarrow \text{Nifty} \\ \leftarrow \text{ETF} \\ \leftarrow \text{bonds} \\ \leftarrow \text{balanced} \end{matrix}$$

Check $AN = 0$	$s_1 = (-1, 1, 0, 0)$	$s_2 = (-0.5, 0, -0.5, 1)$
Budget $(1, 1, 1, 1)$	$-1 + 1 + 0 + 0 = 0 \checkmark$	$-0.5 + 0 - 0.5 + 1 = 0 \checkmark$
Return $(12, 12, 7, 9.5)$	$-12 + 12 + 0 + 0 = 0 \checkmark$	$-6 + 0 - 3.5 + 9.5 = 0 \checkmark$
Exposure $(1, 1, 0, 0.5)$	$-1 + 1 + 0 + 0 = 0 \checkmark$	$-0.5 + 0 + 0 + 0.5 = 0 \checkmark$

Every null space solution

$$x = c s_1 + d s_2 \quad \text{dimension 2: a plane of do-nothing trades}$$

Example: $c = 2, d = 2$ gives $(-2 - 1, 2, -1, 2) = (-3, 2, -1, 2)$. Budget $-3 + 2 - 1 + 2 = 0 \checkmark$, return $-36 + 24 - 7 + 19 = 0 \checkmark$, exposure $-3 + 2 + 0 + 1 = 0 \checkmark$.

Try it: choose c and d

ETF amount $x_2 = c$ Balanced amount $x_4 = d$

Free (you chose): ETF $x_2 = 2$, balanced $x_4 = 2$.

Forced: Nifty $x_1 = -c - 0.5d = -3$, bonds $x_3 = -0.5d = -1$.

Trade $= 2 s_1 + 2 s_2 = (-3, 2, -1, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(-3) + 1(2) + 1(-1) + 1(2) = 0 \checkmark$

Return: $12(-3) + 12(2) + 7(-1) + 9.5(2) = 0 \checkmark$

Exposure: $1(-3) + 1(2) + 0(-1) + 0.5(2) = 0 \checkmark$

Nothing changes: every rule stays at 0, whatever free amounts you pick.

▼ Check yourself (MIT problem 7.1): find R , the rank and the special solutions of $\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix}$.

$A \rightarrow U$: row 3 $- 2 \times$ row 1 $= (0, -12, -3, -21)$; row 3 $+ 3 \times$ row 2 $= (0, 0, 0, 0)$. $U \rightarrow R$: row 2 $\div 4 = (0, 1, \frac{1}{4}, \frac{7}{4})$; row 1 $- 5 \times$ row 2 $= (1, 0, \frac{23}{4}, \frac{1}{4})$. Rank 2; free x_3, x_4 . $s_1 = (-\frac{23}{4}, -\frac{1}{4}, 1, 0)$, $s_2 = (-\frac{1}{4}, -\frac{7}{4}, 0, 1)$. Try it in the lab on the Examples tab.